



Pamantasan ng Lungsod ng Pasig
Alkalde Jose St. Kapasigan, Pasig City
College of Computer Studies



Objectives of the Study

General Objective

The main goal of this study is to develop a Centralized Profiling and Record Authentication System for Senior Citizens in Pasig City. By combining Rule-Based Logic and a Finite State Machine (FSM), the system aims to fix manual delays, lost records, and tracking issues between frontline Barangay Senior Help Desks and the Main OSCA Office.

Specific Objectives

1. **To build a Rule-Based Logic Engine** that automatically checks if an applicant qualifies based on city laws right at the barangay counter.
2. **To create a secure Proxy-Token Generator** using QR codes so that families of bedridden seniors can pre-register online and access a fast-track line.
3. **To design a Finite State Machine (FSM) Workflow Tracker** that monitors the exact status of application folders as they move from the barangay to the main office.
4. **To set up a secure database** with strict user account levels to organize senior records while protecting private details.
5. **To test the system's speed and accuracy** using a simulated environment with data to ensure it is ready for use.

System Functionalities

1. Proxy-Token and Priority Queue Module (Unique Functionality)

- **Pre-Registration:** Relatives of bedridden seniors can upload document photos from home to generate a unique **Proxy QR Code**.
- **Counter Scanning:** Barangay staff scan the QR code using a standard document scanner. The system instantly verifies the relative's identity and places the application into a **High-Priority Line**.



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- **Privacy Guard:** Scanning the QR code with an unauthorized device hides all private details. The data only opens inside the official, secure OSCA system.

2. Localized Compliance Engine (Rule-Based Logic)-ALGORITHM

- **Age Check:** Automatically calculates an applicant's age from their birthdate to ensure they are at least 60 years old before allowing a Senior ID application.
- **Pension Filter:** Screens financial applications by checking if a user's income falls within the allowed local limits and flags anyone already receiving national pensions.
- **Deadline Check:** Automatically counts the days following a resident's passing to strictly enforce local submission windows for burial help, blocking late entries instantly.

3. Batch Tracking (Finite State Machine FSM)-ALGORITHM

- **Workflow Control:** Forces applications to move in a strict, unskippable order through set system stages: **Received-For Review-Verified-Approved-Released.**
- **Error Handling:** Allows main office reviewers to return a batch back to the **Received** state if document scans are blurry, which instantly alerts the barangay staff to rescan the files.
- **History Log:** Automatically logs the exact time, date, and employee account name for every status change to create a clean trail for auditors.

4. Role-Based Access and Security

- **Account Levels:** Limits data access based on job roles. Barangay staff can only view records within their own neighborhood, while main office administrators can view and approve data around Pasig city .
- **Simulated Verification:** Uses a mock connection to simulate verification checks against outside records safely without touching real civilian data during the testing phase.